SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.* (BL2, BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:25**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Saai Vardhan D (192525205) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

Question 1 **10 Marks**

A university computer laboratory allows multiple students to execute programs simultaneously on a shared Linux server.

(a) Identify the type of Operating System suitable for this environment and justify your answer.

(b) State any two responsibilities of the kernel in managing multiple user processes.

(c) Write a C-based system call sequence (or pseudo-code) to create a child process and display the Process IDs of both the parent and child.

Question 2 **10 Marks**

An application needs to store and retrieve student records from a text file.

(a) List the system calls required for file creation, writing, reading, and closing.

(b) Explain the purpose of any two file management system calls.

(c) Write a C-based system call sequence (or pseudo-code) to create a file, write a student name, and close the file.

Question 3 **10 Marks**

A user wants to copy the contents of one file to another.

(a) List the system calls required to complete this operation.

(b) Explain the purpose of any two of these system calls.

(c) Write a pseudo-code or C-based system call sequence to copy the contents of one file to another.

Question 4 **10 Marks**

A program needs to execute another program after creating a child process.

(a) State the purpose of the fork() and exec() system calls.

(b) Explain the role of the wait() system call.

(c) Write a system call sequence using fork(), exec(), and wait().

Question 5 **10 Marks**

An operating system manages multiple files stored on secondary memory.

(a) Explain the importance of file management in an operating system.

(b) State the purpose of the open() and close() system calls.

(c) Write a C-based system call sequence to open a file, read its contents, and close it.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	35%	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25%	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15%	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15%	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10%	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Marks Evaluation:

Question No.	Conceptual Understanding (35%)	Accuracy of Answers (25%)	Application of Knowledge (15%)	Completeness of Response (15%)	Clarity & Presentation (10%)	Total Marks (100%)
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

OPERATING SYSTEMS

CO1 Assessment Test - 1

SHORT ANSWERS

CSA0421

Slot - A

Saai Vardhan. D

192525205

Question - 1

A university computer laboratory allows multiple students to execute programs simultaneously on a shared Linux server

a) Identify the type of Operating System suitable for this environment and justify your answer

A time-Sharing (multi-user, multitasking) Operating System such as Linux. The CPU is divided into small time slices and switched rapidly among users, so every student gets an interactive response whil the server's CPU, memory and I/O are shared efficiently and safely.

b) State any two responsibilities of the Kernel in managing multiple user processes

- Process management: creating, scheduling and terminating processes and allocating CPU time

fairly

- Memory management and protection :- allocating separate memory to each process and preventing one-user's process from accessing another's.

c) Write a C-based system call sequence (or pseudo-code) to create a child process and display the Process ID's of both the parent and child

```
pid = fork ()
if (pid < 0)  print "Fork failed"
else if (pid == 0)
        print "Child PID = ", getpid()
        print "Parent PID = ", getppid()
else                                  // parent
        print "Parent PID = ", getpid()
        print "Child PID = ", pid
        wait (NULL)
```

Question - 2

An application needs to store and retrieve student records from a text file

a) List the system calls required for file creation, writing, reading, and closing.

Creation: creat() or open() with O_CREAT

Writing: write(), Closing: close(),

Reading: read

b) Explain the purpose of any two file management system calls.

- open(): opens (or creates) a file and returns a file descriptor used for all later operations.
- write(): transfers bytes from a program buffer into the file at the current file position.

c) Write a C-based system call sequence (or pseudo-code)

```
fd = open("student.txt", O_CREAT | O_WRONLY, 0644)
if (fd < 0) print "Error"; exit
write(fd, "Saai Vardhan D", length)
close(fd)
```

Question - 3

A user wants to copy the contents of one file to another

a) List the system calls required to another

open() (source and destination), read(), write(), close().

b) Explain the purpose of any two of these system calls

read() :- reads a block of bytes from the source file into a buffer and returns the number of bytes read (0 at end of file)

write() : writes the buffer contents to the destination file.

c) Write a pseudo-code or C-based System call sequence to copy the contents of one file to another

```
Src  = open ("source.txt", O_RDONLY)
dest = open ("dest.txt", O_WRONLY |
                         O_CREAT, 0644)
```

```
while ((n = read (src, buffer, size)) > 0)
        write (dest, buffer, n)
close (src)
close (dest)
```

Question - 4

A program needs to execute another Program after creating a child process.

a) State the propose of the fork() and exec() system calls.

fork(): creates a new child process that is a copy of the parent; returns 0 to the child and the child's PID to the parent.

exec(): replaces the current process image with a new program, which then Stats executing

b) Explain the role of the wait() system call

wait() suspends the parent until a child terminates, collects the child's exit status and lets the OS release the child's resources.

preventing zombie processes.

c) Write a system call sequence using fork(), exec(), and wait()

```
pid = fork()
if (pid == 0)                    // child
    exec("/bin/ls", "ls", "-l")
else if (pid > 0)                // parent
    wait(&status)
    print "Child finished"
else
    print "Fork failed"
```

Question-5

An operating System manages multiple files stored on secondary memory

a) Explain the importance of file management in an operating system

File management organizes data on secondary storage through files and directories. It handles naming, space allocation, access control and protection, sharing between users, and reliable storage

and retrievel, so data is kept safe, organized and easy to find.

b) State the purpose of the open() and close() system calls.

open(): Locates the file, checks permissions and returns a file descriptor for later read/write operations

close(): flushes buffered data, releases the file descriptor and frees the resources held by the open file

c) Write a C-based system call sequence to open a file, read its contents and close it.

```
fd = open("data.txt", O_RDONLY)
if (fd < 0) print "Error"; exit
While ((n=read (fd, buffer, size)) > 0)
        print buffer
close (fd)
```